Keshav Sharma

India | 8168289407 | keshavsharma71982@gmail.com | LinkedIn Link | GitHub Link | Portfolio

EDUCATION

Vellore Institute of Technology — **Chennai, Tamil Nadu**

Bachelor of Technology in Computer Science (Data Science Specialisation) — *Graduation: May 2027*

SKILL SUMMARY

Languages: Python, Java, C++, C, SQL, JavaScript

Software Development: Data Structures & Algorithms, Object-Oriented Programming, Software Design, Problem Solving, Debugging, Testing

Backend & Web: FastAPI, Flask, Node.js, REST APIs, React.js

Databases: SQL, SQLite, MySQL, MongoDB, Redis

Systems & Tools: Git, GitHub, Linux, VS Code, Docker

Machine Learning: Scikit-learn, PyTorch, TensorFlow/Keras, Pandas, NumPy

Engineering Practices: Unit Testing, PyTest, Multithreading, API Development, SDLC, Reproducible Development

EXPERIENCE

VOIS for Tech University Engagement Program **|LINK** — ***Aug – Sep, 2025***

- Engineered a modular data processing pipeline to preprocess, analyze, and query two real-world datasets containing 8,500+ records using LLM-powered natural language interfaces.
- Designed and implemented automated validation and response extraction workflows, improving the consistency and reliability of generated outputs across diverse user queries.
- Optimized data processing by integrating preprocessing, prompt generation, response parsing, and result validation into a reusable workflow, reducing manual analysis effort and improving scalability.

PROJECTS

RedisLite **|LINK**

- Engineered a Redis-inspired in-memory key-value database in Python with a custom RESP parser, multithreaded TCP server, thread-safe storage, O(1) LRU caching, and TTL expiration for concurrent request processing.
- Implemented Write-Ahead Logging (WAL), atomic log compaction, and automatic crash recovery to ensure durable persistence and data consistency under failures and concurrent workloads.
- Developed 46 PyTest tests covering concurrency, edge cases, and recovery scenarios; benchmarked 619K SET ops/sec in-memory and 2.8K ops/sec with WAL persistence to validate reliability and performance.

API Rate Limiter Library **|LINK**

- Built a production-ready Python rate-limiting library implementing five throttling algorithms, Redis integration, and reusable Flask/FastAPI middleware for reliable API traffic control.
- Designed a modular architecture with pluggable storage backends and thread-safe concurrent request handling, enabling flexible deployment across application environments.
- Developed 30+ PyTest tests covering concurrency, edge cases, and algorithm correctness, and packaged the library on PyPI for public installation and reuse.

AI-Powered Expense Receipt Scanner & Analytics Platform **|LINK**

- Built a full-stack Python application using FastAPI, React, SQLAlchemy, and SQLite, integrating EasyOCR, OpenCV, and NLP to automate receipt processing and reduce processing time by 40%.
- Engineered and validated REST APIs with JWT authentication, Google OAuth, CRUD operations, analytics, budget tracking, and data export for reliable application workflows.
- Deployed the application on Render and Vercel, implementing end-to-end workflows across AI processing, backend services, database operations, and frontend validation.

CV-FL: Privacy-Preserving Federated Learning **|LINK**

- Developed a privacy-preserving federated learning framework combining model compression with lightweight verifiable aggregation for resource-constrained edge IoT environments.
- Achieved ~19.7× model compression while maintaining reliable aggregation, and implemented Merkle-tree commitments to detect tampered model updates.
- Evaluated robustness against malicious clients, achieving 100% tamper detection at ≥5% corruption and maintaining reliable operation with up to 30% malicious participants.

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional **|LINK** — **2026**